

Java Program: ArrayIndexOutOfBoundsException Handling

1. Aim

To write a Java program that demonstrates an ArrayIndexOutOfBoundsException by accessing an invalid array index and handles the exception using a try-catch block.

2. Description

An ArrayIndexOutOfBoundsException occurs when a program tries to access an array element using an index that is outside the valid range. In this program, the user enters the array size and its elements. The program then asks for an out-of-bounds index. If the entered index is invalid, Java throws an ArrayIndexOutOfBoundsException. The catch block handles the exception and displays an error message. The finally block closes the Scanner object.

3. Algorithm

1. Start the program.
2. Create a Scanner object to read input from the user.
3. Ask the user to enter the array size.
4. Create an integer array using the entered size.
5. Read and store the array elements.
6. Ask the user to enter an out-of-bounds index.
7. Try to access the array using the entered index.
8. If the index is invalid, catch the ArrayIndexOutOfBoundsException.
9. Display the exception message.
10. Close the Scanner object in the finally block.
11. Stop the program.

4. UML Diagram

ArrayIndexExample	
+ main(String[] args) : void	

5. Source Code with Comments

```
import java.util.Scanner;

public class ArrayIndexExample {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        try {
            // Ask the user for the array size dynamically
            System.out.println("Enter the size of the array: ");
            int size = scanner.nextInt();
            int[] dynamicArray = new int[size];

            // Populate the array dynamically
            System.out.println("Enter " + size + " integers to fill the array:");
            for (int i = 0; i < size; i++) {
                System.out.print("Element at index " + i + ": ");
                dynamicArray[i] = scanner.nextInt();
            }
            // Prompt for an invalid out-of-bounds index
        }
    }
}
```

```

        System.out.println("\n--- Triggering the Exception ---");
        System.out.print("Enter an out-of-bounds index to fetch (e.g., "
            + size + " or higher): ");
        int invalidIndex = scanner.nextInt();

        // Throws ArrayIndexOutOfBoundsException if the index is invalid
        int value = dynamicArray[invalidIndex];
        System.out.println("Value at index " + invalidIndex + " is: " + value);

    } catch (ArrayIndexOutOfBoundsException e) {
        // Handle the exception and display an error message
        System.err.println("\n[ERROR] Caught ArrayIndexOutOfBoundsException!");
        System.err.println("Message: " + e.getMessage()
            + " is not a valid index for this array.");
    } finally {
        // Close the Scanner object
        scanner.close();
    }
}
}

```

6. Compilation and Execution Commands

```

javac ArrayIndexExample.java
java ArrayIndexExample

```

The first command compiles the Java source file. The second command executes the compiled program.

7. Output Screenshot

ArrayIndexExample - Console Output

Enter the size of the array: 3

Enter 3 integers to fill the array:

Element at index 0: 10

Element at index 1: 20

Element at index 2: 30

--- Triggering the Exception ---

Enter an out-of-bounds index to fetch (e.g., 3 or higher): 3

[ERROR] Caught ArrayIndexOutOfBoundsException!

Message: Index 3 out of bounds for length 3 is not a valid index for this array

8. Result

The Java program was successfully executed. It demonstrated an `ArrayIndexOutOfBoundsException` when an invalid array index was accessed, and the exception was successfully handled using a try-catch block. The Scanner resource was closed using the finally block.